

## 1.1 BancsConnect Functionality

BancsConnect supports the following functionality through ISO 8583 messages. The detail ISO 8583 messages have been defined in section 3.

The functional Input data and Output data for each of these messages are as follows

### Inquiries

*Note : All Transactions will receive the A/c Balance as part of the response in field 49 except for Cheque Status Inquiry and Stop payment instruction.*

| #  | FUNCTIONALITY                 | INPUTS                                                                                                                                                                      | OUTPUTS                                                                                                                                                                                                                                                                                                 |
|----|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Balance Inquiry for a A/c     | A/c No                                                                                                                                                                      | Ledger Balance<br>Funds in Clearing<br>Linked A/c Balance<br>Available balance<br>User defined Balance                                                                                                                                                                                                  |
| 2. | Transaction Inquiry for a A/c | A/c No,<br>From Date<br>To Date<br>From Amount<br>To Amount<br>Dr Only<br>Cr Only<br>Inst No<br>Last 'N' Transactions<br>Order<br>(A/D)(Ascending/Descending based on date) | Transaction Date<br>Debit / Credit Indicator<br>Transaction Amount<br>Transaction Particulars                                                                                                                                                                                                           |
| 3. | Mini Statement Inquiry        | A/c No                                                                                                                                                                      | Last 12 Transactions<br>(customizable till 12 )                                                                                                                                                                                                                                                         |
| 4. | Cheque status Inquiry         | A/c No<br>From Chq No<br>To Chq No<br>(Max 50 Cheques)                                                                                                                      | Status of each Chq<br>(Issued,<br>Unused,<br>Passed,<br>Stopped,<br>Destroyed<br>Clearing)                                                                                                                                                                                                              |
| 5. | Customer A/c Inquiry          | Customer ID<br>A/c Type (ALL for All A/c)<br>Number of A/c requested                                                                                                        | All A/c of the Customer or<br>of a particular type<br>requested                                                                                                                                                                                                                                         |
| 6. | Block Inquiry                 | A/c No<br>DCC Id (ALL for ALL<br>Delivery Channels)<br>Reason Code (Optional)<br>Expiry Date (Optional)                                                                     | Last 5 Blocks placed on<br>the A/c containing<br>The reason Code for each<br>block<br>The block Amount                                                                                                                                                                                                  |
| 7. | General A/c Inquiry           | A/c No                                                                                                                                                                      | Customer ID<br>Customer Name<br>A/c Name<br>A/c Open date (value date)<br>A/c Product/Scheme<br>A/c Type<br>(Current/Savings/<br>A/c Status (Frozen,<br>Closed, Dr Only/ Cr Only)<br>Mode of Operation<br>(Single/Joint)<br>A/c Type<br>Joint Holder/s (Max 3)<br>Drawing Power<br>Total Blocked Amount |
| 8. | Term deposit A/c Inquiry      | A/c No                                                                                                                                                                      | Customer ID<br>Customer Name<br>Joint Holders<br>A/c Open Date<br>Tenure (Months &days)<br>Maturity Date<br>A/c Product/Scheme<br>A/c Type (Recurring, Flexi,<br>Fixed)<br>A/c CCY<br>Deposit Amount<br>Maturity Amount<br>Interest rate                                                                |

|     |                       |                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|-----------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.  | Loan A/c Inquiry      | A/c No                                                                                    | Customer ID<br>Customer Name<br>Joint Holders<br>Loan A/c open date<br>Loan Tenure (Months & days)<br>Loan Product<br>Loan A/c Type (Term/EMI)<br>Loan Sanctioned Amount<br>Loan Disbursed Amount<br>Installment Amount<br>Next Installment due date<br>Overdue Amount                                                                                                                                                                                                   |
| 10. | Currency rate Inquiry | From CCY<br>To CCY<br>Type (TC/ Cash etc)<br>Incase amount is to be converted, the Amount | <b>Fixed currency code</b><br><b>Fixed currency units -</b><br><b>Variable currency code -</b><br><b>Variable currency units -</b><br><b>Converted amount -</b><br>(Will be zero if transaction amount is not present<br>amount fields will have sign ("+") followed by sixteen character amount with implied decimal based on reference currency code.) (eg: 0000000005000000)<br><b>Flag indicating whether the rate is latest or not –</b><br>Y/N a per business date |

## Requests

|     |                             |                                                                         |                                                                                                                   |
|-----|-----------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 11. | Cheque Book Issue request   | A/c No<br>No of leaves (3 Char)<br>Design Type (5 Char)                 | Status (accepted/Rejected)<br>To be processed at back-end<br>using Workflow<br>Reference ID for later Inquiry     |
| 12. | Statement Of A/c request    | A/c No                                                                  | Status<br>(Accepted/ Rejected)<br>To be processed at back-end<br>using Workflow<br>Reference ID for later Inquiry |
| 13. | DD Issue request            | From A/c<br>DD Amt<br>DD CCY<br>Beneficiary Name<br>Payable at Location | Status<br>(Accepted/ Rejected)<br>To be processed at back-end<br>using Workflow<br>Reference ID for later Inquiry |
| 14. | Cheque Deposit Notification | A/c<br>Txn Amount<br>Txn CCY<br>Cheque No                               | Status (Accepted Rejected).<br>To be processed at back-end<br>using Workflow<br>Reference ID for later Inquiry    |
| 15. | Cash Deposit Notification   | A/c<br>Txn Amount<br>Txn CCY                                            | Status (accepted,<br>Rejected)<br>To be processed at back-end<br>using Workflow<br>Reference ID for later Inquiry |
| 16. | Request Status Inquiry      | A/c No<br>Request Reference No                                          | Status of Request                                                                                                 |

## Transactions

|     |                                                                                                                                                                 |                                                                                                    |                                                                                            |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 17. | Cash Withdrawal                                                                                                                                                 | A/c<br>Txn Amount<br>Txn CCY<br>Fees<br>Fees CCY                                                   | Success/ Reject<br>Balance After<br>Withdrawal                                             |
| 18. | POS Debit Authorization request                                                                                                                                 | A/c<br>Txn amount<br>Txn CCY<br>Fees<br>Fees CCY                                                   | Success / Reject<br>Balance after debit<br>authorization.                                  |
| 19. | Cash deposit authorization request                                                                                                                              | A/c<br>Amount<br>Txn CCY<br>Fees<br>Fees CCY                                                       | Success / Reject<br>(Accepted for<br>verification)<br>Balance after debit<br>authorization |
| 20. | Debit Authorization request                                                                                                                                     | A/c<br>Amount<br>Txn CCY<br>Fees<br>Fees CCY                                                       | Success / Failure<br>Balance after debit<br>authorization                                  |
| 21. | Credit Authorization request                                                                                                                                    | A/c<br>Amount<br>Txn CCY<br>Fees<br>Fees CCY                                                       | Success / Failure<br>Balance after credit<br>authorization                                 |
| 22. | Funds Transfer between linked A/c                                                                                                                               | From A/c<br>To A/c<br>Txn Amount<br>Txn CCY<br>Fees<br>Fees CCY                                    | Success/ Reject<br>Balance After Debit and<br>credit for both A/c                          |
| 23. | Third party funds transfer                                                                                                                                      | From A/c<br>To A/c<br>Txn Amount<br>Txn CCY<br>Fees<br>Fees CCY                                    | Success/ Reject<br>Balance After Debit only<br>for debit A/c                               |
| 24. | Payment Transaction request                                                                                                                                     | From A/c<br>Utility Code/To A/c<br>Txn Amt<br>Txn CCY<br>Fees<br>Fees CCY                          | Success/ Reject<br>Balance After Debit                                                     |
| 25. | Remittances<br>(Type of Payment Transaction with<br>extra details )                                                                                             | From A/c<br>To A/c No<br>To Bank<br>To Branch<br>Txn Amt<br>Txn CCY<br>Fees<br>Fees CCY            | Success/ Reject<br>Balance After Debit                                                     |
| 26. | Modify Block funds request<br>(Add/Remove)                                                                                                                      | A/c<br>Block Amount<br>Block ID<br>Block Reason Code<br>Remarks<br>Block Expiry Date               | Success / Failure<br>A/c Status<br>Balance after Block                                     |
| 27. | Unblock and debit funds                                                                                                                                         | A/c No<br>Amount to be debited<br>Block ID to be unblocked<br>Block Reason code<br>Unblock Remarks | Success / Failure<br>A/c Status<br>Balance after debit                                     |
| 28. | DD Issue authorization request with<br>channel specific charges This assumes<br>that the DD is being issued and printed<br>from the Delivery Channel (E.g. ATM) | From A/c<br>DD Amt<br>DD CCY<br>DD Fees<br>Fees CCY<br>Beneficiary Name<br>Payable at Location     | Success /failure                                                                           |
| 29. | Cheque-book Issued authorization<br>request                                                                                                                     | A/c No<br>Start Cheque No<br>Number of leaves                                                      | Success- Failure                                                                           |

|     |                                     |                                                            |                                               |
|-----|-------------------------------------|------------------------------------------------------------|-----------------------------------------------|
| 30. | Stop Payment Instruction            | A/c No<br>Start Leaf No<br>No of Leaves<br>Reason Code     | Success / failure<br>If Failure , Reason Code |
| 31. | Reversal for all above Transactions | A/c<br>Reversal Amount<br>Original Transaction<br>elements | Has to be success<br>always                   |

## 2 ISO 8583 interface to BancsConnect

### 2.1 List of messages in version 2.0

#### Existing Messages

1. Repeat messages/advice
2. Network Messages (Logon, Echo test)
3. Cash withdrawal authorization request
4. Cash withdrawal authorization reversal
5. Dr. authorization request
6. Dr request reversal
7. Cr. Authorization request
8. Cr request reversal
9. Funds Transfer authorization request
10. Funds Transfer request reversal
11. Utility Payments authorization request
12. Utility Payments reversal
13. Balance Inquiry
14. Mini Statement Inquiry
15. POS purchase authorization request –
16. POS Purchase authorization request reversal
17. Stop Check instruction
18. Stop Check Instruction reversal
19. Check status Inquiry
20. User defined Messages
21. Miscellaneous Messages – Used by the bank to extend BancsConnect functionality

#### New Messages in Version 2.0

22. Cash deposits Notification - The appropriate 'workflow' needs to be developed by the bank in FiNACLE to handle this request.
23. Cheque deposits Notification - The appropriate 'workflow' needs to be developed by the bank in FiNACLE to handle this request.
24. Cheque book request - The appropriate 'workflow' needs to be developed by the bank in FiNACLE to handle this request.
25. Statement of accounts request - The appropriate 'workflow' needs to be developed by the bank in FiNACLE to handle this request.
26. DD Issue request - The appropriate 'workflow' needs to be developed by the bank in FiNACLE to handle this request. *This message will be used only when FiNACLE has to issue the DD.*
27. Hold /Release Funds request
28. Block Inquiry Message
29. Hold/Release Funds request reversal
30. Dr against held funds authorization request
31. Dr against held funds authorization request reversal
32. Cash deposits authorization request reversal
33. Issue DD authorization request – *Will be used by front end applications that can actually issue DD's.*
34. Issue DD authorization request reversal
35. Issue Checkbook authorization request - *Will be used by front end applications that can actually issue*
36. Issue Checkbook authorization request reversal
37. General Account Inquiry
38. Loan Account Inquiry
39. Deposit Account Inquiry
40. Currency Rate Inquiry
41. Request Status Inquiry

*Cash withdrawal, Cash deposit and POS transactions where transaction currency is a foreign currency but the customer account is in local currency are also supported.*

## 2.2 Communication Protocol

1. Communication protocol between the Delivery channel Controller and BancsConnect will be TCP/IP.
2. BancsConnect will service all messages (transaction messages and the echo-test message) at a specified port.
3. BancsConnect supports the following models for connection management:
  - a. Open and close a connection for each message
  - b. Maintain a pool of connections and process one message at a time (i.e. send request, wait and receive response) on each of the connections
  - c. Maintain a pool of connections and process multiple messages at a time on each of the connections. In this case the delivery of responses are not guaranteed to be in the same order as the corresponding requests.

**Note:**

If BancsConnect receives any field, not specified in this document, within an incoming request or advice, BancsConnect will ignore it. Also, 'reversals' should be originated by the DCC only if the request to BancsConnect times out. In cases where BancsConnect responds with an 'error response', BancsConnect would take the responsibility of 'reversing' the transaction in FiNACLE, if required. Similarly 'repeats' should be originated by the DCC only if the request to BancsConnect times out.

## 2.3 Security

For ISO 8583 messages, as of now, the only way to totally secure the messages transmitted between the delivery channel controller and BancsConnect is to use router-based access control and encryption techniques. However BancsConnect has an in-built IP address-based access control mechanism to prevent unauthorized messages to BancsConnect. Also, the 'customization' features of BancsConnect, allows a bank to implement its own authentication/checksum mechanisms to prevent unauthorized / tampering of ISO 8583 messages to BancsConnect.

## 2.4 Message Formats

### 2.4.1 Message headers

|                        |    |                                                                                        |
|------------------------|----|----------------------------------------------------------------------------------------|
| Version number         | N1 | 1 = ISO 8583:1993                                                                      |
| Message class          | N1 | 1 = Authorization<br>2 = Financial<br>3 = File requests<br>4 = Reversal<br>8 = Network |
| Message function       | N1 | 0 = Request<br>1 = Request Response<br>2 = Advice<br>3 = Advice Response               |
| Transaction originator | N1 | 0 = Acquirer<br>1 = Acquirer Repeat<br>4 = Other<br>5 = Other Repeat                   |
| Primary bit map        | B8 | Set depending upon the primary fields used                                             |

## 2.4.2 Messages sent or Received by BancsConnect

| Message | Description                                       | Originator   | Destination  |
|---------|---------------------------------------------------|--------------|--------------|
| 1200    | Request                                           | DCC          | BancsConnect |
| 1201    | Request Repeat                                    | DCC          | BancsConnect |
| 1210    | Request Response                                  | BancsConnect | DCC          |
| 1220    | Advice (Transactions authorized locally)          | DCC          | BancsConnect |
| 1221    | Advice repeat (Transactions authorized locally)   | DCC          | BancsConnect |
| 1230    | Advice response (Transactions authorized locally) | BancsConnect | DCC          |
|         |                                                   |              |              |
| 1420    | Reversal advice                                   | DCC          | BancsConnect |
| 1421    | Reversal advice repeat                            | DCC          | BancsConnect |
| 1430    | Reversal advice response                          | BancsConnect | DCC          |
| 1804    | Network Management Request (Logon/Logoff) – N/A   |              |              |
| 1814    | Network Management Response (Logon/Logoff) – N/A  |              |              |
| 1804    | Network Management Request (Echo Test)            | DCC          | BancsConnect |
| 1814    | Network Management Response (Echo Test)           | BancsConnect | DCC          |

### 2.4.3 Network Management Messages

| Bit | Description                                             | 1804 | 1814 | Format   | Remarks                                                                                                                                                                                                                                  |
|-----|---------------------------------------------------------|------|------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Secondary bitmap                                        | M    | M    | B8       | Bitmap corresponding to the secondary fields used.                                                                                                                                                                                       |
| 11  | System Trace Audit Number                               | M    | ME   | Anp12    | ISO supports N6. BancsConnect will accept Anp12 format for requests received by it.                                                                                                                                                      |
| 12  | Local Transmission time                                 | M    | ME   | N14      | Time in YYYYMMDDhhmmss format. ISO supports YYMMDDhhmmss.                                                                                                                                                                                |
| 24  | Function Code                                           | M    | ME   | N3       | Please refer to function codes table.                                                                                                                                                                                                    |
| 39  | Action Code                                             | -    | M    | N3       | BancsConnect will return 800 (OK) for echo-test requests and 902 (Invalid transaction) for all other requests.                                                                                                                           |
| 59  | Transport Data                                          | M    | ME   | Ans..999 | For requests received by BancsConnect, this field will be echoed by BancsConnect within the response.                                                                                                                                    |
| 93  | Transaction destination Institution identification code | M    | ME   | N..11    | For requests received by BancsConnect, this field will be echoed by BancsConnect within the response.                                                                                                                                    |
| 94  | Transaction originator Institution identification code  | M    | ME   | N..11    | For requests received by BancsConnect, this field will be echoed by BancsConnect within the response.                                                                                                                                    |
| 123 | Delivery channel controller Id                          | M    | ME   | A3       | Three character code identifying the controller. Hard-coded value for each controller e.g. "SWT": Switch, "TLB": Tele-banking, "FAB": <a href="#">FINACLE@Branch</a> etc. This field will be echoed by BancsConnect within the response. |

## 2.4.4 Transaction messages Structure

| Bit | Description                                | 1200 | 1210 | 1420 | 1430 | Format | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|--------------------------------------------|------|------|------|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Secondary bit map                          | M    | M    | M    | M    | B8     | Set depending upon bits used in secondary bitmap.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2   | Primary Account number/customer id         | C    | CE   | C    | CE   | N..19  | Either this field or extended PAN (Field 34) would be available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 3   | Processing code                            | M    | ME   | M    | ME   | N6     | Since BancsConnect uses only the account identifications (Fields 102 and 103), account types given in this field will be ignored. BancsConnect will echo the entire field on responses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4   | Transaction amount                         | M    | ME   | M    | ME   | N16    | <p>Transaction Amount for all debit and credit transactions e.g.</p> <ul style="list-style-type: none"> <li>• Cash Withdrawals</li> <li>• Funds transferred</li> <li>• POS Purchase Amount</li> <li>• The new block amount for Modify Block request. It is unsigned. The sign information is sent in ISO field 125. The total block amount on the account (ISO field 102) is updated with this amount.</li> <li>• The amount to be unblocked and debited for Unblock Debit request.</li> <li>• The transaction amount for normal debit/credit</li> <li>• DD Amount</li> <li>• Remittance amount</li> <li>• Payment amount</li> </ul> <p>For reversal transactions, this field indicates the amount being reversed. ISO has N12.</p> |
| 5   | Amount, Reconciliation                     | O    | OE   | O    | OE   | N16    | Funds to be transferred between acquirer and the issuer. ISO has N12. This field is not used by FINACLE, but could be used by bank's own scripted logic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 9   | Conversion rate, reconciliation            | C    | CE   | C    | CE   | N8     | Required if amount, reconciliation (Field 5) is provided. This field is not used by FINACLE, but could be used by bank's own scripted logic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 11  | System Trace Audit Number                  | M    | ME   | M    | ME   | Anp12  | This number will remain the same for multiple messages within the same transaction. ISO supports N6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 12  | Local Transaction Date & Time              | M    | ME   | M    | ME   | N14    | YYYYMMDDhhmmss (ISO supports two digit year). This is the time at card acceptor's location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 15  | Date, settlement                           | O    | OE   | O    | OE   | N8     | Date in YYYYMMDD format for transferring funds from card issuer to acquirer. This field is not used by FINACLE, but could be used by bank's own scripted logic. ISO supports YYMMDD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 16  | Date, Conversion                           | C    | CE   | C    | CE   | N8     | Date in YYYYMMDD format. Required if amount, reconciliation has been provided. ISO format is MMDD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 17  | Capture Date                               | M    | ME   | M    | ME   | N8     | Date in YYYYMMDD format. This will be the business date of the transaction. ISO format is MMDD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 24  | Function Code                              | M    | -    | M    | -    | N3     | Please refer to function codes table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 30  | Original Amounts                           | -    | -    | C    | -    | N32    | Original amount in case of partial reversals. Original amount in case of pre-authorization completion transactions. ISO supports N24.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 32  | Acquiring Institution Identification Code  | M    | ME   | M    | ME   | N..11  | Code identifying the acquirer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 33  | Forwarding Institution Identification Code | O    | OE   | O    | OE   | N..11  | Can be put when forwarding institution is not the same as institution originating the message. Will be echoed by BancsConnect. This field is not used by FINACLE, but could be used by bank's own scripted logic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 34  | Extended Primary Account number            | C    | CE   | C    | CE   | Ns..28 | Can be used in if Primary Account Number (Field 2) is not provided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Bit | Description                  | 1200 | 1210 | 1420 | 1430 | Format   | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|------------------------------|------|------|------|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 37  | Retrieval Reference Number   | O    | OE   | O    | OE   | Anp12    | In cases where the DCC does not maintain the original STAN, this will be used by BC as the STAN for all messages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 39  | Action Code                  | -    | M    | -    | M    | N3       | Action codes that can be returned by BancsConnect. These will be as per table specified below.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 41  | Card Acceptor Terminal Id    | O    | OE   | O    | OE   | Ans16    | Device Id which could be Terminal Id in case of ATM or POS, Channel Id in case of tele-banking. This can be used to derive the Device Cash A/C. ISO supports 8 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 42  | Card Acceptor Identification | O    | OE   | O    | OE   | Ans15    | To derive Device Cash A/C from field 41, this field will be used to find whether transaction originated from a device directly connected to the DCC..                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 43  | Card Acceptor Name/Location  | O    | -    | O    | -    | Ans..99  | The Location of the Card Acceptor terminal Used for logging/custom reports.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 46  | Amounts, Fees                | O    | OE   | O    | OE   | Ans..300 | Up to six fees that need to be posted to cardholder account. Format of the field is:<br>1. Fee Type Code N2<br>2. Currency Code, Fee N3 or A3<br>3. Amount of Fee X + N16 where X is the D (Debit) or C (Credit) (Should generally be debit for requests and credit for reversals)<br>4. Conversion rate, fee, N8<br>5. Amount for reconciliation X+N16<br>6. Currency for reconciliation N3<br>(FINACLE currently supports only five fees)<br>This field will contain the fees used for all transactions where fees will be collected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 48  | Additional Data, Private     |      | C    | -    | C    | Ans..999 | For all transactions other than Block Inquiry, Cheque Status inquiry and stop Payment Request BancsConnect will send:<br>1. 17 char ledger balance (Source account)<br>2. 17 char available balance<br>3. 17 char float balance<br>4. 17 char FFD balance<br>5. 17 char user defined balance<br>6. 3 char balance currency code<br>7. 14 char Fallback time in YYYYMMDDhhmmss format, (Blank if data center is not in fallback mode)<br><br>In case of Own A/c funds Transfers the credit A/c balances will also be sent in the same format of the debit A/c balances<br><br>Destination account balance will be specified only in case of transfer transaction. All the balance fields will have sign ("+" or "-") followed by sixteen character amount with implied decimal based on balance currency code. It will be filled up only for authorized transactions.<br>-----<br>For Block Inquiry, Cheque Status Inquiry and Stop Payment request please refer to the specific messages in section 4.6 |
| 49  | Transaction currency Code    | M    | ME   | M    | ME   | A3 or N3 | BancsConnect supports all ISO CCY codes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 50  | Currency, Reconciliation     | C    | CE   | C    | CE   | A3 or N3 | If amount, reconciliation (Field 5) is provided, this field is required. This field is not used by FINACLE, but could be used by bank's own scripted logic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Bit | Description                                            | 1200 | 1210 | 1420 | 1430 | Format   | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|--------------------------------------------------------|------|------|------|------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 56  | Original Data Elements                                 | C    | -    | M    | -    | N..43    | Required for reversals and pre-authorization completions. Consists of:<br>1. 4 char Message Type Identifier<br>2. 12 char System Trace Audit No (ISO supports 6) or RRN of the original message where DCC does not maintain STANS.<br>3. 14 char Local Date and Time (ISO supports 12) of the original Transaction<br>4. 11 char Acquiring Institution Id (2 char length will precede institution id)                                                          |
| 59  | Transport Data                                         | O    | OE   | O    | OE   | Ans..999 | Transport data, if provided within the original, will be echoed unchanged by BancsConnect. This field is not used by FiNACLE, but could be used by bank's own scripted logic..                                                                                                                                                                                                                                                                                 |
| 62  | Reserved for Private Use                               | C    | CE   | C    | CE   | Ans..999 | This field is used by the following messages<br>• Cheque Status Inquiry<br>• Stop Payment Request<br>• Cheque Book Requests<br>• Funds Transfers/Cash withdrawals using Cheques<br>Please refer the specific messages in section 4.6                                                                                                                                                                                                                           |
| 63  | Reserved for Private Use                               | O    | -    | O    | -    | Ans..28  | This field is used for Utility Payment Transactions. Please refer to the specific message in section X.X                                                                                                                                                                                                                                                                                                                                                       |
| 66  | Amounts, original fees                                 | -    | -    | C    | -    | Ans..300 | Required in case of partial reversal, if the fees have changed. Up to six fees that represent the fees originally posted to cardholder account. Format of the field is:<br>1. Fee Type Code N2<br>2. Currency Code, Fee N3 or A3<br>3. Amount of Fee X + N16 where X is the debit or credit indicator<br>4. Conversion rate, fee, N8<br>5. Amount for reconciliation X+N16<br>6. Currency for reconciliation N3<br>(FiNACLE currently supports only five fees) |
| 94  | Transaction Originator Institution Identification Code | O    | OE   | O    | OE   | N..11    | If specified in reversal advice message, BancsConnect will echo it within its response. This field is not used by FiNACLE, but could be used by bank's own scripted logic.                                                                                                                                                                                                                                                                                     |
| 99  | Settlement Institution Identification Code             | O    | -    | O    | -    | N..11    | Financial institution at which the accounts are held by the parties settling. This field is not used by FiNACLE, but could be used by bank's own scripted logic..                                                                                                                                                                                                                                                                                              |
| 102 | Account Identification-1                               | M    | -    | M    | -    | Ans..38  | For inquiry and debits:<br>1. 11 char Bank Id (AN)<br>2. 8 char SOL id (Branch code) (AN)<br>3. 19 char a/c number (AN)<br>(ISO allows Ans..28)                                                                                                                                                                                                                                                                                                                |
| 103 | Account Identification-2                               | C    | -    | C    | -    | Ans..40  | For credits:<br>1. 2 Char Code (If credit account number is fully specified, use spaces. Otherwise this would be used to derive the actual account number from the fields below. It could be, for example the option selected on ATM)<br>2. 11 Char Bank Id<br>3. 8 char SOL id<br>4. 19 char a/c number<br>(ISO allows ANS..28)                                                                                                                               |

| Bit | Description                    | 1200 | 1210 | 1420 | 1430 | Format   | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|--------------------------------|------|------|------|------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 123 | Delivery channel controller Id | M    | ME   | M    | ME   | A3       | Three character code identifying the controller.<br>Hard-coded value for each controller by the controller<br>e.g. "SWT": Switch,<br>"TLB": Telebanking,<br>"FAB": <a href="#">FiNACLE@Branch</a>                                                                                                                                                                                                                                                   |
| 124 | Terminal Type                  | O    | OE   | O    | OE   | A3       | This field is not used by but could be used by bank's own scripted logic.<br>Possible values could be:<br>1. POS<br>2. ATM<br>3. TLB (Telebanking)<br>4. ITB (Internet banking)<br>5. TLR (Teller)                                                                                                                                                                                                                                                  |
| 125 | Reserved for private use       | C    | C    | -    | -    | Ans..999 | Please refer to the specific messages defined for the following in section 4.6<br>1. General A/c Inquiry<br>2. Customer A/c Inquiry<br>3. Transaction Inquiry<br>4. Mini statement inquiry<br>5. Statement request<br>6. DD Issue authorization<br>7. DD Issue request<br>8. Remittances<br>9. Modify Block<br>10. Unblock Debit<br>11. Currency rate Inquiry<br>12. Request Status Inquiry<br>13. Deposit A/c Inquiry<br>14. Loan A/c Inquiry<br>. |
| 126 | Reserved for private use       | -    | C    | -    | -    | Ans..999 | Same as field 125.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 127 | Reserved for private use       | -    | C    | -    | -    | Ans..999 | Same as field 125                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## 2.5 Processing Codes for Functionality Supported

| Code | Meaning                     | Remarks                                                                                                                                                                                                                                            |
|------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00   | Normal Purchase (POS)       | Account 1 (Field 102) will be debited online                                                                                                                                                                                                       |
| 01   | Cash Withdrawal             | Account 1 (Field 102) will be debited online. If field 123 is 'FAB' <a href="#">FiNACLE@Branch</a> , then cash account of user-id specified in field 41 is credited .If check number is specified in field 62 that check will be marked as 'used'. |
| 18   | Debit request               | Account in field 102 will be debited by the transaction amount.                                                                                                                                                                                    |
| 21   | Cash Deposit Notification   | BancsConnect will pass the request to the host to be processed by the host through custom workflow                                                                                                                                                 |
| 24   | Cheque Deposit Notification | BancsConnect will pass the request to the host to be processed by the host through custom workflow.                                                                                                                                                |
| 28   | Credit request              | Credit the account specified in Field103.                                                                                                                                                                                                          |
| 31   | Balance Inquiry             | Available, ledger and float balances would be returned in Field 48                                                                                                                                                                                 |

|    |                          |                                                                                                                                                                                                                                                                                                                                                                                       |
|----|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 36 | Currency rate inquiry    | The from currency code ,to currency code between which conversion is to be done and the rate code to be used for the conversion will be specified in the field 125. The rate will be returned in field 125 as 12 character field with decimal.                                                                                                                                        |
| 37 | Customer account inquiry | Fetches all open accounts of the specified account type for the given customer. Maximum of 150 accounts will be fetched.                                                                                                                                                                                                                                                              |
| 38 | Mini Statement Inquiry   | Up to last 5 transactions for account 1 (Field 102) will be returned in Field 125                                                                                                                                                                                                                                                                                                     |
| 39 | Block inquiry message    | For block inquiry message, BancsConnect sends the details of the latest 5 blocks on the account in field 48 with reason code.                                                                                                                                                                                                                                                         |
| 40 | Funds Transfer           | Funds transferred from account 1 (Field 102) to account 2 (Field 103). If check number is specified in field 62 that check will be marked as 'used'                                                                                                                                                                                                                                   |
| 48 | Transfer debit           | Funds debited from account 1 (Field 102). If check number is specified in field 62 that check will be marked as 'used'                                                                                                                                                                                                                                                                |
| 49 | Transfer credit          | Funds credited to account 2 (Field 103)                                                                                                                                                                                                                                                                                                                                               |
| 50 | Payment                  | Funds transferred from account 1 (Field 102) to account derived from account 2 (Field 103). The same code will be used for Payment to parties holding accounts with other banks. In that case account 2 will belong to another bank.<br>In case the Account is specified in this field the account specified will be verified and funds will be transferred to the account specified. |

|    |                                                                              |                                                                                                                                                                                                                                                                                                                                                            |
|----|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17 | Modify Block/Unblock /Unblock and debit/<br>(refer to field 125 for details) | This message will block/ unblock /Unblock debit the mentioned amount in the message from the A/c specified in field 102 If the amount is negative the funds will be unblocked. In case of a positive amount the funds will be blocked.                                                                                                                     |
| 10 | DD Issue authorization                                                       | Account 1 (field 102) is debited for the total amount(field 4 + field 46) and a DD issue record is created for amount in field 4 with data provided in field 125. This message is for DD's that will be issued either at the Branch or at Call centers. The main point here is that the inventory being used here is from the delivery channels inventory. |
| 10 | DD issue Request                                                             | Account 1 (field 102), amount in field 4 , and DD Payable at (field 125 – Place) will be available in 'script hook' in FINACLE to spawn a custom workflow                                                                                                                                                                                                  |

| Code | Meaning                         | Remarks                                                                                                                                                                                                                                                                                                               |
|------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 81   | Cheque book Issue authorization | Account 1 (field 102) is debited for the charges ( field 46) and a Checkbook issue record is created for Account 1(field 102) with data provided in field 62. Similar to DD Issue authorization, this request will be sent by DCC's where the DCC itself is issuing the cheque book and not from a central Inventory. |
| 82   | Account inquiry                 | This message will provide basic details of the A/c such as A/ holder's name/s<br>A/c Balances, A/c open date and A/c Status                                                                                                                                                                                           |
| 83   | Loan account inquiry            | This message will provide Loan A/c details such as Loan A/c holders Name/s,<br>Sanctioned amount, Interest rate, Disbursed amount, Pending Amount, Installment Amount and next Installment date                                                                                                                       |
| 84   | Term deposit Inquiry            | This message will provide Deposit A/c details such as Deposit A/c Holders Name/s, Deposit Amount, Deposit open date, Deposit Maturity date, Interest date and Deposit Maturity Amount.                                                                                                                                |
| 90   | Check Book Request              | Account (field 102) and Number of cheque leaves (field 62) will be available in 'script hook' in FiNACLE to spawn a custom workflow                                                                                                                                                                                   |
| 91   | Cash deposits authorization     | Valid only if field 123 is ' <a href="#">FiNACLE@Branch</a> '. Account 2 (field 103) is credited online and cash account of user-id specified in field 41 is debited online.                                                                                                                                          |
| 92   | Statement Request               | Account 1 (field 102) and date range (field 125) will be available in 'script hook' in FiNACLE to spawn a custom workflow                                                                                                                                                                                             |

| Code | Meaning                       | Remarks                                                                                                                                                                                                                                                                                                                         |
|------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 93   | Transaction Inquiry           | BancsConnect will return back list of statement in fields(125,126,and 127) for the Account (field 102) and the given date range and continuation transaction id (field 126).                                                                                                                                                    |
| 94   | Cheque Status Inquiry         | Status of cheques issued to account 1 (Field 102) will be returned in Field 125.                                                                                                                                                                                                                                                |
| 95   | Stop Cheque Instruction       | Cheques specified in Field 62 issued to account 1 (Field 102) will be marked as 'stopped'                                                                                                                                                                                                                                       |
| 96   | Pay away messages             | Used by Bill Payment and Presentment Product "PayAway"                                                                                                                                                                                                                                                                          |
| 97   | Miscellaneous processing code | Miscellaneous messages are messages are private messages between FiNACLE and the respective delivery channel. This message uses the private fields 125 to 127 of the ISO message structure for data transmission. The format of these fields for each private message is pre-defined by both Infosys and the respective vendor. |
| 98   | Request Status Inquiry        | This message is for receiving the status of a request such as DD request, cash deposit request, Cheque Deposit request messages that would be processed in FiNACLE using customized work flows. This message is not supported in SIS.                                                                                           |

## 2.6 Messages

### 2.6.1 Inquiries

#### 2.6.1.1 Balance Inquiry

Input:

| Bit | Description     | REMARKS                                        |
|-----|-----------------|------------------------------------------------|
| 3   | Processing code | Valid value : 31                               |
| 102 | Account number  | Account for which balance inquiry is required. |

Output:

| Bit | Description             | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes (Refer section 4.4.7)                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 48  | Additional data private | Balances of account as given below:<br>8. 17 char ledger balance (Source account)<br>9. 17 char available balance<br>10. 17 char float balance<br>11. 17 char FFD balance<br>12. 17 char user defined balance<br>13. 3 char balance currency code<br>14. 14 char Fallback time in YYYYMMDDhhmmss format, (Blank if data center is not in fallback mode)<br><br>All the balance fields will have sign (“+” or “-”) followed by sixteen character amount with implied decimal based on balance currency code. |

### 2.6.1.2 Mini Statement Inquiry

**Input:**

| Bit | Description     | REMARKS                                       |
|-----|-----------------|-----------------------------------------------|
| 3   | Processing code | Valid value : 38                              |
| 102 | Account number  | Account for which mini statement is required. |

**Output:**

| Bit | Description             | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes Refer section 4.4.7                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 48  | Additional data private | Balances of the account. Refer section 4.4.4                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 125 | Reserved field          | The transaction details are sent in the format given below. *<br><br><ol style="list-style-type: none"><li>1. 8 char date in YYYYMMDD</li><li>2. 8 char Instrument Id (right justified, left padded with spaces)</li><li>3. 5 char tran report code (Right padded with spaces)</li><li>4. 40 char Tran Particulars (Right padded with spaces)</li><li>5. 1 char Db/Cr Indicator</li><li>6. 17 char Formatted Amount (with explicit decimal) (left padded with zeros)</li></ol> |

*\*Mini-statement format is changed from the previous BancsConnect format. It can be formatted according to the requirement of delivery channels in the BancsConnect script.*

## 2.6.1.3 Transaction Details Inquiry

## Input:

| Bit | Description              | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code          | Valid value : 93                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 102 | Account number           | Account for which transaction details are required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 125 | Reserved for private use | <p>This is the criteria filed for transaction inquiry</p> <p><b>Start Date</b> – 8 char date in YYYYMMDD ( Mandatory )</p> <p><b>End Date</b> - 8 char date in YYYYMMDD (Optional, by default BOD Date)</p> <p><b>Low Amount</b> - 17char amount with decimal</p> <p><b>High Amount</b> - 17char amount with decimal</p> <p><b>Begin Cheque Num</b> - 8 char begin cheque number</p> <p><b>End Cheque Num</b> - 8 char end cheque number</p> <p><b>Number of Records</b> - 2 char ( 00 to 20 ) (Mandatory)</p> <p><b>Sorting Order</b>- 1 char 'A'scending/'D'escending (Mandatory)</p> <p><b>Credit or Debit</b> – 'C' for credit only, 'D' for debit only, 'B' for both debit and credit. (Mandatory)*</p> <p>The transaction inquiry can fetch a maximum of 20 records at a time. For the next set of records, the following should be populated from the previous record.</p> <p><b>Last Transaction Date</b>- 8 char in YYYYMMDD</p> <p><b>Last Transaction Id</b> - 9 char Last transaction id (right justified left padded with spaces)</p> <p><b>Last part tran number</b> - 4 char Last part tran number (right justified left padded with spaces)</p> <p><b>Last transaction posted date time</b> - 14(91-104)char last posted date time in YYYYMMDDHHMISS</p> <p><b>Last balance</b> - 17 char last balance with decimal</p> |

\* **Credit or Debit** flag in the criteria is a new addition.

## Output:

| Bit | Description             | REMARKS                                      |
|-----|-------------------------|----------------------------------------------|
| 39  | Response code           | Valid ISO response codes Refer section 4.4.7 |
| 48  | Additional data private | Balances of the account. Refer section 4.4.4 |

|     |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 125 | Reserved field | <p>This field gives the details of the transactions</p> <p><b>More data flag</b> : 1 char ('Y'/'N') This indicates whether there are more records for the given criteria.</p> <p><b>Number of statements</b> : 2char (00 to 20) Number of statements in this fetch</p> <p>The following are repeated as many number of statements</p> <p><b>Transaction date</b> : 8 char ( YYYYMMDD )</p> <p><b>Transaction Id</b> : 9 char (right justified, left padded with spaces)</p> <p><b>Part tran serial num</b> : 4 char (right justified left padded with spaces)</p> <p><b>Tran Type</b> : 1 char ( C - Cash, T - Transfer, L - Clearing )</p> <p><b>Tran sub type</b> : 2 char ( BI, CI, NP, NR)</p> <p><b>Debit credit indicator</b> : 1 char ( D - Debit, C - Credit )</p> <p><b>Tran value date</b> : 8 char ( YYYYMMDD )</p> <p><b>Transaction Amount</b> : 17 char ( with decimal )</p> <p><b>Transaction particulars</b> : 50 Char (left justified, right padded with spaces)</p> <p><b>Transaction Posted date</b> : 14 char ( YYYYMMDDHHMISS )</p> <p><b>Instrument number</b> : 8 char (right justified, left padded with spaces)</p> <p><b>Balance at the end of the transaction</b> : 17 char ( with decimal )</p> |
|-----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 2.6.1.4 Cheque status Inquiry

**Input:**

| Bit | Description     | REMARKS                                               |
|-----|-----------------|-------------------------------------------------------|
| 3   | Processing code | Valid value : 94                                      |
| 62  | Cheque Details  | 1. Start Cheque Number N8<br>2. No of leaves N3       |
| 102 | Account number  | Account for which cheque status inquiry is required.. |

**Output:**

| Bit | Description             | REMARKS                                                                                                                                                                                                                                                                                                                                                        |
|-----|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes (Refer section 4.4.7)                                                                                                                                                                                                                                                                                                                 |
| 48  | Additional data private | Balances of the account. (Refer section 4.4.4)                                                                                                                                                                                                                                                                                                                 |
| 125 | Reserved field          | For cheque status inquiry, it will return one character cheque status as follows:<br>1. Paid i.e. cheque used ('P')<br>2. Cautioned i.e. a warning set up on the cheque ('C')<br>3. Destroyed when account is being closed ('D')<br>4. Stopped ('S')<br>5. Paid cheque returned ('R')<br>6. Unused ('U')<br>7. Sent to user but not acknowledged by user ('I') |

## 2.6.1.5 Customer account inquiry

## Input:

| Bit | Description     | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code | Valid value : 37                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 125 | Reserved field  | <b>Page number requested</b> 2char numeric (01 for first page, LP for last.)<br><b>Number of records required in a page</b> 2char numeric (maximum 25)<br><b>Customer Id</b> (Finacle customer id) 9 char right justified, left padded with spaces<br><b>account type</b> 3 char<br>(ALL – for all types of accounts,<br>SBA – SB a/c<br>CCA – Cash credit accounts<br>CAA – Current accounts<br>ODA – Overdraft accounts<br>LAA – Loan accounts<br>TDA – Term deposit a/c)<br>If the account type is not specified, 'ALL' will be assumed. |

## Output:

| Bit | Description              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code            | Valid ISO response codes. Refer section 4.4.7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 125 | Reserved for private use | <b>Page number fetched</b> 2char (01 for the first page. <b>LP</b> for the last page. Maximum 99 pages can be displayed. Each page can contain a maximum of 25 records)<br><b>Number of records in this page</b> 2char numeric (maximum 25)<br><b>Customer Id</b> (Finacle customer id) 9 char right justified, left padded with spaces<br><b>Account type requested</b> 3 char (valid values as given above)<br><br>Maximum of 50 records of the account number and account type<br><b>Account number</b> 16 char right padded with space<br><b>Account type</b> 3 char<br><b>Clear Balance</b> 17 char (with sign (“+” or “-”) followed by sixteen character amount with implied decimal based on account currency code )<br><b>Account currency code</b> 3 char |

### 2.6.1.6 General Account Inquiry

#### Input:

| Bit | Description     | REMARKS                                 |
|-----|-----------------|-----------------------------------------|
| 3   | Processing code | Valid value : 82                        |
| 102 | Account number  | Details of this account will be fetched |

#### Output:

| Bit | Description              | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code            | Valid ISO response codes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 48  | Additional data private  | 17 char ledger balance<br>17 char available balance<br>17 char funds in clearing<br>17 char FFD balance (If the account belongs to a pool, this will indicate the Term deposit contribution to the pool)<br>17 char user defined balance<br><br><b>All the balance fields will have sign (“+” or “-”) followed by sixteen character amount with implied decimal based on balance currency code. It will be filled up only for authorized transactions.</b>                                                                                                                                                                                                                                                                                                                                                                                                           |
| 125 | Reserved for private use | For General A/c Inquiry This field will contain the following details<br>Customer Id (9 Char right justified, left padded with spaces)<br>Customer Name (80 char left justified right padded with spaces)<br>Account Name (80 char left justified right padded with spaces)<br>A/c open date YYYYMMDD (8 Char)<br>A/c Product (Scheme code) 5 Char left justified right padded with spaces<br>A/c Type – 3 Char (Valid value : SBA/CCA/ODA/CAA)<br>A/c Status – 1 Char (F- Frozen, C - Closed, D - Debit frozen, R - Credit Frozen, M – Dormant, A – Active,O – Open, I - Inactive)<br>Mode of Operation code -5char (left justified right padded with spaces)<br>Joint holder Name1(80 Char)<br>Joint holder Name2 (80 Char)<br>Joint holder Name3 (80 Char)<br>GL subhead code (5 char)<br>Account sol id (8 char)<br>17 char Drawing power<br>17 char Lien amount |

## 2.6.1.7 Loan Account Inquiry

## Input:

| Bit | Description     | Remarks                                      |
|-----|-----------------|----------------------------------------------|
| 3   | Processing code | Valid value : 83                             |
| 102 | Account number  | Details of this loan account will be fetched |

## Output:

| Bit | Description              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code            | Valid ISO codes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 48  | Additional data private  | Account balances (Refer:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 125 | Reserved for private use | Customer Id (9 Char right justified, left padded with spaces)<br>Customer Name (80 char left justified right padded with spaces)<br>A/c Open Date – YYYYMMDD (8 Char)<br>Loan period months – 3char left padded with zeros (eg: 036)<br>Loan period days – 3 char left padded with zeros (eg: 090)<br>Loan product (Scheme code) –5 Char<br>Interest Rate – 10 Char with decimal<br>Loan Type – 'F'ixed, 'E'MI (1 Char)<br>Next installment due date (YYYYMMDD) – 8 Char<br>Joint Holders name1 (80 Char)<br>Joint Holders name2 (80 Char)<br>Joint Holders name3 (80 Char)<br>Sanctioned Amount<br>Disbursed Amount<br>Installment Amount<br>Over due amount<br>(All the amount fields will have sign (“+”) followed by sixteen character amount with implied decimal based on Loan Account currency code.) (eg: +0000000005000000)<br>Loan Account currency – 3 char |

## 2.6.1.8 Deposit Account Inquiry

## Input:

| Bit | Description     | Remarks                                              |
|-----|-----------------|------------------------------------------------------|
| 3   | Processing code | Valid value : 84                                     |
| 102 | Account number  | Details of this term deposit account will be fetched |

## Output:

| Bit | Description              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code            | Valid ISO response codes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 48  | Additional data private  | Account balances (Refer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 125 | Reserved for private use | Term deposit account details will be populated in this field as given below.<br>Customer Id (9 Char right justified, left padded with spaces)<br>Customer Name (40 char left justified right padded with spaces)<br>A/c Open Date – YYYYMMDD (8 Char)<br>Deposit period months – 3char left padded with zeros (eg: 036)<br>Deposit period days – 3 char left padded with zeros (eg: 090)<br>A/c Maturity Date -YYYYMMDD (8 Char)<br>A/c Type – (1 Char) 'R'ecurring, Fle'X'i, 'F'ixed , 'C'ertificate of deposit, 'N' otice deposit<br>A/c Product(Scheme code) – 5 Char (right padded with spaces)<br>Joint holder Name 1(80 Char) (right padded with spaces)<br>Joint holder Name 1(80 Char) (right padded with spaces)<br>Joint holder Name 1(80 Char) (right padded with spaces)<br>Interest Rate – 10 Char with decimal (left padded with zeros)<br>Deposit Amount<br>Maturity Amount<br>(All the amount fields will have sign (“+”) followed by sixteen character amount with implied decimal based on Deposit Account currency code.) (eg: +0000000005000000)<br>Account currency code - 3 char |

## 2.6.1.9 Block Inquiry:

## Input:

| Bit | Description     | REMARKS                                                                                                                                                                                                                                                                                                                                                         |
|-----|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code | Valid value : 39                                                                                                                                                                                                                                                                                                                                                |
| 102 | Account number  | <ul style="list-style-type: none"><li>Account on which block inquiry has to be performed.</li></ul>                                                                                                                                                                                                                                                             |
| 125 | Reserved field  | <ul style="list-style-type: none"><li>The following criteria should be given for fetching the lien details</li></ul> <ol style="list-style-type: none"><li>DCC Id : 3 char (ALL for all delivery channels)</li><li>Lien reason code: 5 character (left justified, right padded with spaces) (Optional)</li><li>Lien expiry date : YYYYMMDD (Optional)</li></ol> |

## Output:

| Bit | Description             | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response Code           | Refer response code table for valid response codes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 48  | Additional data priavte | For block inquiry message, BancsConnect sends the details of the latest 5 blocks on the account in this field. The following is the format<br>5 char reason code 1<br>17 char total Block Amount for reason code 1<br>5 char reason code 2<br>17 char total Block Amount for reason code 2<br>5 char reason code 3<br>17 char total Block Amount for reason code 3<br>5 char reason code 4<br>17 char total Block Amount for reason code 4<br>5 char reason code 5<br>17 char total Block Amount for reason code 5                                                                                                                                                           |
| 125 |                         | In Modify Block/Unblock Debit/Block Inquiry response messages, this field indicates the status of the account if the account is closed/frozen : <ul style="list-style-type: none"><li>Account status indicator : 2 char<br/>TF – Total Freeze<br/>DF – Debit Freeze<br/>AC – Account Closed</li></ul> In all other cases it will be 00. <ul style="list-style-type: none"><li>Total lien amount on the account by the DCC. (If the DCC id is “ALL”, the total lien amount on the account will be given)</li></ul> (The amount fields will have sign (“+”) followed by sixteen character amount with implied decimal based on Account currency code.) (eg: +0000000005000000) |

## 2.6.1.10 Currency Rate Inquiry

| Bit | Description              | Remarks                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code          | 36                                                                                                                                                                                                                                                                                                                                                                                              |
| 125 | Reserved for private use | From CCY (Transaction currency code) (3 Char)<br>To CCY (Reference currency code). (3 Char)<br>[Note: <i>This will be the account currency code in case the inquiry is being done for a transaction where transaction currency is different from account currency</i> ]<br>Rate Type (5 char) TC, CSH, BILL etc as specified in FiNACLE rate code.<br>Transaction amount :17 character amount.. |

## Output:

| Bit | Description              | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code            | Valid ISO response codes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 125 | Reserved for private use | <b>Fixed currency code</b> - 3 char<br><b>Fixed currency units</b> - (12 char rate with decimal, left padded with zeros)<br><b>Variable currency code</b> - 3 char<br><b>Variable currency units</b> - (12 char rate with decimal, left padded with zeros)<br><b>Converted amount</b> -<br>(Will be zero if transaction amount is not present amount fields will have sign (“+”) followed by sixteen character amount with implied decimal based on reference currency code.) (eg: 0000000005000000)<br><b>Flag indicating whether the rate is latest or not</b> – 1 char ( <b>Y</b> – if rate is latest, <b>N</b> – if rate is not latest). The rate will be compared against the business date (capture date ) field.* |

*\*Flag indicating whether the rate is latest or not is introduced in the latest version.*

## 2.6.2 Transactions

### 2.6.2.1 Cash withdrawal request

**Input:**

| Bit | Description          | REMARKS                                                 |
|-----|----------------------|---------------------------------------------------------|
| 3   | Processing code      | Valid value : 01                                        |
| 4   | Transaction amount   | Amount to be debited (Refer section 4.4.4)              |
| 46  | Amount fees          | Charges (Refer section 4.4.4)                           |
| 49  | Transaction currency | Transaction currency code. All valid ISO currency codes |
| 102 | Account number       | Account from which amount is being withdrawn            |

**Output:**

| Bit | Description             | REMARKS                                                    |
|-----|-------------------------|------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes (Refer section 4.4.7)             |
| 48  | Additional data private | Balances of the account after debit. (Refer section 4.4.4) |

### 2.6.2.2 Funds Transfer authorization request

**Input:**

| Bit | Description          | REMARKS                                                 |
|-----|----------------------|---------------------------------------------------------|
| 3   | Processing code      | Valid value : 40                                        |
| 4   | Transaction amount   | Amount to be transferred between accounts               |
| 46  | Amount fees          | 5 types of Fees (Refer section 4.4.4)                   |
| 49  | Transaction currency | Transaction currency code. All valid ISO currency codes |
| 102 | Account number - 1   | Debit account number (Source account)                   |
| 103 | Account number - 2   | Credit account number (Destination account)             |

**Output:**

| Bit | Description             | REMARKS                                                                                                           |
|-----|-------------------------|-------------------------------------------------------------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes (Refer section 4.4.7)                                                                    |
| 48  | Additional data private | Balances of the account after debit followed by balances of the credit account after credit (Refer section 4.4.4) |

### 2.6.2.3 Stop Check Instruction

**Input:**

| Bit | Description     | REMARKS                                                                                                   |
|-----|-----------------|-----------------------------------------------------------------------------------------------------------|
| 3   | Processing code | Valid value : 95                                                                                          |
| 46  | Amount fees     | 5 types of Fees (refer section 4.4.4)                                                                     |
| 62  | Cheque Details  | For cheque book request<br>1. Cheque start number N8<br>2. Number of Leaves N3<br>3. Stop Reason code An5 |
| 102 | Account number  | Account for which cheque needs to be stopped.                                                             |

**Output:**

| Bit | Description             | REMARKS                                                                                                                                |
|-----|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes (Refer section 4.4.7)                                                                                         |
| 48  | Additional data private | Balances of the account. (Refer section 4.4.4)                                                                                         |
| 125 | Reserved field          | This field will contain the status of the cheque leaves after trying stop request. The status is same as sent in check status inquiry. |

#### 2.6.2.4 Utility Payments authorization request

**Input:**

| Bit | Description          | REMARKS                                                                   |
|-----|----------------------|---------------------------------------------------------------------------|
| 3   | Processing code      | Valid value : 50                                                          |
| 4   | Transaction amount   | Amount to be transferred between accounts (Refer section 4.4.4)           |
| 46  | Amount fees          | 5 types of Fees (Refer section 4.4.4)                                     |
| 49  | Transaction currency | Transaction currency code. All valid ISO currency codes                   |
| 102 | Account number - 1   | Debit account number (Refer section 4.4.4)                                |
| 103 | Account number - 2   | 2 character payee code or<br>Credit account number. (Refer section 4.4.4) |

**Output:**

| Bit | Description             | REMARKS                                                         |
|-----|-------------------------|-----------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes (Refer section 4.4.7)                  |
| 48  | Additional data private | Balances of the debit account after debit (Refer section 4.4.7) |

## 2.6.2.5 Modify Block request:

## Input:

| Bit | Description                    | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code                | Valid value : 17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4   | Transaction Amount             | <ul style="list-style-type: none"> <li>New block Amount for Modify Block Request</li> <li>It is unsigned.</li> <li>Sign Info is available in Field 125</li> <li>Total block amount in the account (Field 102) is updated with this amount.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 102 | Account number                 | <ul style="list-style-type: none"> <li>Account on which modify block has to be performed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 125 | Reserved Field for private use | <ul style="list-style-type: none"> <li>1 char sign indicator<br/>Valid values               <ul style="list-style-type: none"> <li>a) P (Positive) : it indicates that the amount given in field 4 (transaction amount) is the additional block amount on the account given in Field 102.</li> <li>b) N (Negative) : Indicates that the account (field 102) needs to be unblocked to the extent of the amount specified in transaction amount filed.</li> </ul> </li> <li>25 char Block Id (Should be left justified and padded with spaces. <i>If this field is blank, the block id will be formed as Dccid/Reasoncode. For eg: if the reason code is SHARE and the dcc id is ATM, the block id will be ATM/SHARE</i>)</li> <li>5 char reason code (left justified)</li> <li>50 char remarks (e.g., transaction reference number of the brokerage system)</li> <li>8 char Block Expiry Date in YYYYMMDD format. If not present, default Expiry Date is assumed to be 20991231</li> </ul> |

## Output:

| Bit | Description   | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code | <p>Refer response code table for valid response codes;<br/>In addition to the normal response codes, the following response codes are added</p> <p><b>184</b> : Requested Block operation failed since Account is closed/frozen. Account status is sent in ISO field 125. Note that brokerage requests can be processed on a Credit freeze account and not permitted on Total Freeze/Debit freeze/Closed Accounts.</p> <p><b>185</b> : Invalid Currency/Transaction Amount</p> <p><b>186</b> : Block Does Not Exist</p> |

| Bit | Description             | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response code           | <p>Refer response code table for valid response codes;<br/>In addition to the normal response codes, the following response codes are added</p> <p><b>184</b> : Requested Block operation failed since Account is closed/frozen. Account status is sent in ISO field 125. Note that brokerage requests can be processed on a Credit freeze account and not permitted on Total Freeze/Debit freeze/Closed Accounts.<br/><b>185</b> : Invalid Currency/Transaction Amount<br/><b>186</b> : Block Does Not Exist</p>                 |
| 48  | Additional data private | <p><b>5 different balances of the source account after modify block will be sent.</b><br/>For all transactions other than cheque inquiry and stop cheque requests, BancsConnect will send:</p> <p>15. 17 char ledger balance (Source account)<br/>16. 17 char available balance<br/>17. 17 char float balance<br/>18. 17 char FFD balance<br/>19. 17 char user defined balance<br/>20. 3 char balance currency code<br/><b>14 char Fallback time in YYYYMMDDhhmmss format, (Blank if data center is not in fallback mode)</b></p> |
| 125 |                         | <p>In Modify Block/Unblock Debit/Block Inquiry response messages, this field indicates the status of the account if the account is closed/frozen :</p> <ul style="list-style-type: none"><li>• TF – Total Freeze</li><li>• DF – Debit Freeze</li><li>• AC – Account Closed</li></ul> <p>In all other cases it will be 00.</p>                                                                                                                                                                                                     |

## 2.6.2.6 Unblock Debit request :

## Input:

| Bit | Description              | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code          | Valid value : 18                                                                                                                                                                                                                                                                                                                                                                                           |
| 4   | Transaction Amount       | <ul style="list-style-type: none"> <li>Amount to be unblocked and debited from the account.</li> <li>It is unsigned.</li> <li>Sign Info is available in Field 125</li> <li>Total block amount in the account (Field 102) is updated with this amount.</li> </ul>                                                                                                                                           |
| 102 | Account number           | <ul style="list-style-type: none"> <li>Account on which block needs to be removed and debit transaction is to be created.</li> </ul>                                                                                                                                                                                                                                                                       |
| 125 | Reserved for private use | <ul style="list-style-type: none"> <li>1 char sign indicator. Always P.</li> <li>25 char Block Id (Left justified, filled with spaces. If this field is blank, then the block id will be taken as "DCCId/Reason code")</li> <li>5 char reason code</li> <li>8 char Block expiry date (in YYYYMMDD format)</li> <li>50 char remarks (e.g., transaction reference number of the brokerage system)</li> </ul> |

## Output :

| Bit | Description             | REMARKS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39  | Response Code           | Refer response code table for valid response codes                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 48  | Additional data private | <p>5 different balances of the source account after unblock debit will be sent.<br/>For all transactions other than cheque inquiry and stop cheque requests, BancsConnect will send:</p> <p>21. 17 char ledger balance (Source account)<br/>22. 17 char available balance<br/>23. 17 char float balance<br/>24. 17 char FFD balance<br/>25. 17 char user defined balance<br/>26. 3 char balance currency code<br/>14 char Fallback time in YYYYMMDDhhmmss format, (Blank if data center is not in fallback mode)</p> |
| 125 |                         | <p>In Modify Block/Unblock Debit/Block Inquiry response messages, this field indicates the status of the account if the account is closed/frozen :</p> <ul style="list-style-type: none"> <li>TF – Total Freeze</li> <li>DF – Debit Freeze</li> <li>AC – Account Closed</li> </ul> <p>In all other cases it will be 00.</p>                                                                                                                                                                                          |

## 2.6.3 Requests

### 2.6.3.1 Cash Deposit notification request :

#### Input:

| Bit | Description               | REMARKS                                 |
|-----|---------------------------|-----------------------------------------|
| 3   | Processing code           | Valid value : 21                        |
| 4   | Transaction amount        | The deposit amount                      |
| 49  | Transaction currency code | All valid ISO currency codes            |
| 103 | Account number            | Account to which cash is to be credited |

#### Output:

| Bit | Description              | REMARKS                                                            |
|-----|--------------------------|--------------------------------------------------------------------|
| 39  | Response code            | Valid ISO response codes listed in table -                         |
| 48  | Additional data private  | Balances of the account.                                           |
| 125 | Reserved for private use | Can be used to send reference id (generated by the custom program) |

**2.6.3.2 Cheque Deposit notification request:****Input:**

| Bit | Description        | REMARKS                                                                                                                        |
|-----|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code    | Valid value : 24                                                                                                               |
| 4   | Transaction amount | Amount of the cheque                                                                                                           |
| 62  | Cheque details     | 8 characters long cheque number<br>(This field can be customized to hold multiple cheque numbers, bank code, branch code etc.) |
| 103 | Account number     |                                                                                                                                |

**Output:**

| Bit | Description             | REMARKS                                                           |
|-----|-------------------------|-------------------------------------------------------------------|
| 39  | Response code           | Standard ISO response codes                                       |
| 48  | Additional data private | Balances of the account.                                          |
| 125 | Reserved field          | Reference Id generated in the custom program can be sent in this. |

**2.6.3.3 Cheque book request :**

| Bit | Description     | REMARKS                                                                                                     |
|-----|-----------------|-------------------------------------------------------------------------------------------------------------|
| 3   | Processing code | Valid value : 90                                                                                            |
| 62  | Cheque Details  | For cheque book request<br>1. Number of leaves - N3<br>2. Design type - AN5 (Can be used for customization) |
| 102 | Account number  | Account for which cheque book is to be issued.                                                              |

**Output:**

| Bit | Description             | REMARKS                                                             |
|-----|-------------------------|---------------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes                                            |
| 48  | Additional data private | Balances of the account.                                            |
| 125 | Reserved field          | Reference Id generated by custom program can be sent in this field. |

#### 2.6.3.4 Statement of account request:

**Input:**

| Bit | Description     | REMARKS                                                                                                                                     |
|-----|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code | Valid value : 92                                                                                                                            |
| 102 | Account number  | Account number for which statement is to be processed.                                                                                      |
| 125 | Reserved Field  | For Statement request the field can optionally contain the data required for statement generation. This can be decided while customization. |

**Output:**

| Bit | Description             | REMARKS                                                                |
|-----|-------------------------|------------------------------------------------------------------------|
| 39  | Response code           | Valid ISO response codes                                               |
| 48  | Additional data private | Balances of the account.                                               |
| 125 | Reserved field          | Reference Id generated by the custom program can be sent in this field |

## 2.6.3.5 DD Issue Request :

**Input:**

| Bit | Description          | REMARKS                                                                                                                                                                                                                                                                                                               |
|-----|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | Processing code      | Valid value : 98                                                                                                                                                                                                                                                                                                      |
| 4   | Transaction amount   | DD amount                                                                                                                                                                                                                                                                                                             |
| 49  | Transaction currency | All valid ISO currency codes                                                                                                                                                                                                                                                                                          |
| 102 | Account number       | Account number from which the dd amount is to be debited.                                                                                                                                                                                                                                                             |
| 125 | Reserved Field       | For DD Issue request the field will be formatted as follows:<br>1. DD Payable at Bank *(In Case of Correspondent Banks) ( 8 Char )<br>2. DD payable at branch code (8 char). Code should match branch code set-up in FiNACLE.<br>3. Beneficiary Name (40 Char )<br>* Code should match branch code set-up in FiNACLE. |

**Output:**

| Bit | Description             | REMARKS                                      |
|-----|-------------------------|----------------------------------------------|
| 39  | Response code           |                                              |
| 48  | Additional data private | Balances of the account.                     |
| 125 | Reserve Field           | Reference Id generated by the custom program |

## 2.6.4 Function Codes

| Code | Meaning                                                               |
|------|-----------------------------------------------------------------------|
| 200  | Original Authorization request or advice                              |
| 400  | Reversal                                                              |
| 801  | Logon – Not supported since BancsConnect handles stand-in processing  |
| 802  | Logoff – Not supported since BancsConnect handles stand-in processing |
| 831  | Echo-test (Received by BancsConnect)                                  |

## 2.6.5 Action codes returned by BancsConnect

| Code | Meaning                                                                                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 000  | Financial transaction has been approved (DCC should go ahead and complete the transaction)                                                                                                                                                             |
| 114  | Invalid account number (DCC should reject the transaction)                                                                                                                                                                                             |
| 115  | Requested function not supported (First two digits of processing code or Function code is invalid) (DCC should reject the transaction)                                                                                                                 |
| 116  | Insufficient funds (DCC should reject the transaction)                                                                                                                                                                                                 |
| 119  | Transaction not permitted to card holder (DCC should reject the transaction)                                                                                                                                                                           |
| 121  | Withdrawal amount limit exceeded. (DCC should reject the transaction)                                                                                                                                                                                  |
| 180  | Transfer Limit Exceeded (DCC should reject the transaction)                                                                                                                                                                                            |
| 800  | Network message was accepted                                                                                                                                                                                                                           |
| 902  | Invalid transaction (Invalid function code within network management messages) (DCC should reject the transaction)                                                                                                                                     |
| 904  | Format Error (Any format related errors etc) (DCC should reject the transaction)                                                                                                                                                                       |
| 906  | Cut-over in progress (When Stand-in Server is handing over control to FiNACLE. (DCC should reject the transaction and try again after some time or do local authorization)                                                                             |
| 907  | Card issuer inoperative (When BancsConnect cannot contact FiNACLE and cannot do stand-in processing.) (DCC should reject transaction and do local authorization )                                                                                      |
| 909  | System malfunction (Sent by BancsConnect for errors like database corrupted) (DCC should reject transaction and do local authorization)                                                                                                                |
| 911  | Card issuer timed out (BancsConnect contacts FiNACLE and FiNACLE times out and BancsConnect cannot do stand-in processing) (DCC should mark transaction as 'suspect' and do local authorization )                                                      |
| 913  | Duplicate transmission. This error is returned if a duplicate message is received and the transaction amount is different. The DCC will have to treat this transaction as 'suspect' and carry out the necessary steps to process suspect transactions. |

### Note:

*If the DCC times out on its request to BancsConnect or receives a '000' response after its timeout value then it should mark that transaction as 'suspect' and send a 'reversal' message to BancsConnect.  
In any case if the DCC does local authorization, it should be done under a new STAN.*

### 2.6.6 Notations used

|      |                                                                                                                                                                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A    | Alphabetic characters A through Z and a through z                                                                                                                                                                                                            |
| N    | Numeric digits 0 through 9                                                                                                                                                                                                                                   |
| P    | Pad character i.e. space                                                                                                                                                                                                                                     |
| S    | Special Characters                                                                                                                                                                                                                                           |
| An   | Alphabetic and Special Characters                                                                                                                                                                                                                            |
| Ns   | Numeric and Special Characters                                                                                                                                                                                                                               |
| Anp  | Alphabetic, Numeric and Pad (Space) characters                                                                                                                                                                                                               |
| Ans  | Alphabetic, Numeric and Special Characters                                                                                                                                                                                                                   |
| MM   | Month, 01 through 12                                                                                                                                                                                                                                         |
| DD   | Day, 01 through 31                                                                                                                                                                                                                                           |
| YY   | Year, 00 through 99                                                                                                                                                                                                                                          |
| YYYY | Year, 0000 through 9999                                                                                                                                                                                                                                      |
| Hh   | Hour, 00 through 23                                                                                                                                                                                                                                          |
| Mm   | Minute, 00 through 59                                                                                                                                                                                                                                        |
| Ss   | Second, 00 through 59                                                                                                                                                                                                                                        |
| B    | Binary representation of data                                                                                                                                                                                                                                |
| ..nn | Variable length data up to nn characters. There will be two or three character length (depending upon whether maximum data length is 99 or 999) at the beginning of the element to identify the number of positions following to the end of the data element |
| DCC  | Delivery Channel Controller                                                                                                                                                                                                                                  |
| SAF  | Store and Forward File                                                                                                                                                                                                                                       |
| PBF  | Positive Balance File                                                                                                                                                                                                                                        |
| SOL  | Service Outlet                                                                                                                                                                                                                                               |

### 2.6.7 Conventions used for denoting field presence

|    |                                                                                                        |
|----|--------------------------------------------------------------------------------------------------------|
| M  | Field must be present                                                                                  |
| ME | If field is present in request, the same value has to be echoed in the response                        |
| -  | If the field is present in request, it will be ignored. It will not be put by BancsConnect in response |
| C  | Field is mandatory under certain conditions                                                            |
| CE | If the field has been provided in the request, it will be echoed by BancsConnect within response       |
| O  | Field is optional within the request. If present, BancsConnect will take certain action on it.         |
| OE | If the field is present within request, it will be echoed by BancsConnect within response              |

## 3 Delivery Channel Management Application (DCMA)

### 3.1 Introduction

Apart from the online messaging interface to FiNACLE via BancsConnect (as described above), there needs to be a batch interface between the DCC and FiNACLE as well. The reasons are:

1. How does the bank ensure that all dr./cr. requests and reversals originated from the DCC have been effected in FiNACLE and effected correctly.

One way to do this is to assume that both the DCC and FiNACLE applications behave perfectly under normal conditions and problems could arise only when there is a problem in the connection between the DCC and BancsConnect (like timeouts). So manually, check the 'suspect' transactions in the DCC log at the end of each day, to ensure correct operations. Also, periodically take 'summary transaction reports' from FiNACLE and the DCC logs to check if they tally.

Another way is to automate this process by the DCC supplying transactions originated by it and FiNACLE supplying transactions effected by it to a central agency who will try and match the two and come out with the discrepancies so that the bank can take action. This kind of strategy guards against software errors in the DCC or FiNACLE applications also. FiNACLE's FMR (Inter Branch reconciliation module) has the capability to do this kind of matching provided it is supplied with originating messages from the DCC and responding messages from FiNACLE.

2. If the DCC does local authorizations and is not able to 'Replay' the SAF to FiNACLE because of link issues for an extended period (which could happen if a bank deploys FiNACLE in a flexi-centralization mode), then it can use the FMR (or BORPI) mechanism (which is a batch file transfer based interface) to get the transactions effected in FiNACLE.
3. If the bank decides to delegate responsibility of 'ATM cash accounting' of ATMs 'owned' by different branches, there is an issue with 'not-on-us' cash withdrawals in which the 'cash withdrawal' message needs to be sent to both the branches (one in which the debit account resides and the other which owns the ATM where the cash withdrawal takes place). Some ATM switches (like the Interlink switch) provides for 'LORO' messages to take care of this issue while most switches do not. Hence FiNACLE provides for a batch interface (BORPI) which allows 'cash messages' from the ATMs to be effected in bulk into their respective branch's cash account.
4. For 'us-on-network' transactions processed by the ATM switch, no message is typically sent online by the switch to FiNACLE. However, the bank still needs to recover the transaction amount (for cash withdrawals) and charges from the settlement institution. Hence FiNACLE provides for a batch interface (BORPI) which allows these settlement transactions to be uploaded.
5. The bank may need to apportion the transaction charges to various accounts/branches for revenue accounting. This can either be done online (using the script hooks in FiNACLE) or it can be done in batch. Hence FiNACLE provides for a batch interface (BORPI) which allows these revenue accounting transactions to be uploaded.
6. For POS purchases, only the Customer account is debited online. The merchant owning the POS should receive the credit. This is done in batch. Hence FiNACLE provides for a batch interface (BORPI) which allows the merchant's account to be credited.

Hence, we envisage that with each DCC there is a corresponding DCMA that processes the transaction log and creates the necessary files for the FMR and BORPI interface of FiNACLE, typically once a day. This application processes the transaction log file created by delivery channel controller. Batch interface files generated by this application are fairly generic and bank can decide the actual account numbers used and so on. Please note that the DCC vendor should either provide these applications or the bank needs to develop this in-house since they are very DCC dependent.

### 3.2 Inputs to DCMA

1. Transaction Log File (TLF) which contains:
  - (a) Transaction requests or advices successfully posted to FiNACLE
  - (b) Advice within the Store And Forward (SAF) file pending to be posted to FiNACLE
  - (c) Suspect transaction requests or advices such as those for which response was not received within timed out period

### 3.3 Outputs from DCMA

1. FMR file (For Inter-branch Reconciliation Module)
2. BORPI files (For posting transactions using FiNACLE™ Open Request Processing Interface)
3. Reports on cash and cheque deposits that are to be processed on the basis of the transaction log file.

### 3.4 FMR File

Processing Logic for Finacle Message Reconciliation (FMR) file

1. Only "us-on-us" transactions should be considered for generation of FMR file. "Network-on-us" and "us-on-network" transactions should not be put within the FMR file. (There should be a logic in the BancsConnect script-hook to specify that for 'network-on-us' transactions, FiNACLE should not send a responding message to FMR).
2. Only transactions that affect customer accounts should be processed. This means that cash withdrawal, transfers, payments, POS. purchases, DD issue authorizations and Debit against held funds should be processed. Other transactions like balance inquiry, statement inquiry, check inquiry, stop check request, deposits, checkbook reorder request, statement request, hold/release funds should not be processed.

3. Transfers and Utility Payments should result in two records in FMR file. One record will be for the debit leg and the other will be for the credit leg. All other transactions should result only in one record within the FMR file. (In case of payments to account holders in other banks, credit will be to a collection account that belongs to the bank).
4. Please note that FMR file will basically be a text file with each record separated by linefeed characters.

### 3.4.1 Header Record

| Record Type            | Char(2) | Mandatory | "80"                                                                                                                                                                                                                  |
|------------------------|---------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Destination FMR Center | Char(2) | Mandatory | FMR Center to which the file is to be sent.<br>FINACLE allows the bank to set up multiple FMR centers for load balancing. Each of them is given a 2-character code. For Central FMR, this field should be set to "BC" |